CS 440 Assignment 1

Fast Trajectory Replanning in unknown gridworlds

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A. Replanning, head to head

expansions, trajectory and replans per world, h = manhattan, larger-g ties

	expansions	moves	replans	solved
scatter 101x101				
Repeated Forward A*	3,353	216	56	50/50
Repeated Backward A*	**26,885**	212	57	50/50
Adaptive A*	**3,215**	215	56	50/50
maze 51x51				
Repeated Forward A*	26,002	540	207	50/50
Repeated Backward A*	**58,396**	581	217	50/50
Adaptive A*	**13,345**	559	211	50/50

B. Tie-breaking (Part 3)

Repeated Forward A*, equal-f ties broken toward smaller or larger g

	smaller g	larger g	small / large
scatter	44,696	3,394	**13.2x**
maze	45,799	15,058	**3.0x**

C. Heuristics, one search, known map

expansions for a single A* search from the agent to the target

	manhattan	differential	yours
scatter	942	933	**575**
maze	3,811	891	**882**

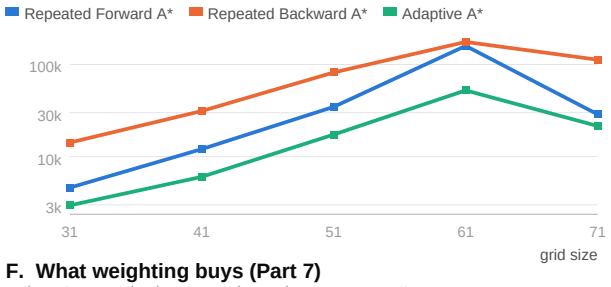
D. Landmark staleness (Part 6c)

how far the tables may drift before rebuilding, in the maze loop

rebuild after	expansions	seconds	probes
age > 1	8,866	0.66	1,332,855
age > 4	9,349	0.41	742,524
age > 16	10,594	0.27	260,890
age > 64	11,278	0.09	66,272
never rebuild	21,519	0.10	11,243

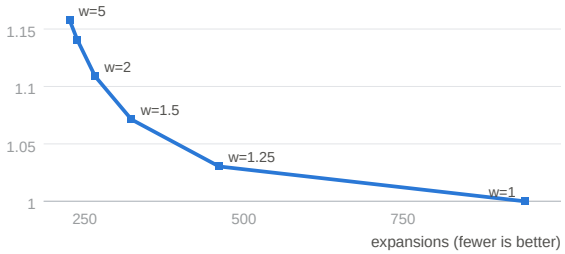
E. Cost against grid size

mean A* expansions per world, maze family, log scale

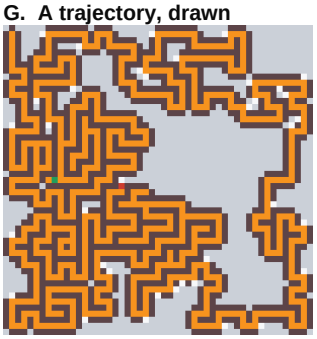


F. What weighting buys (Part 7)

path cost over optimal, one search on a known map, scatter



G. A trajectory, drawn



Adaptive A* on maze seed 40553. Orange is the route actually walked, dark cells are blockages it discovered, flat grey is terrain it never observed.

Q1 Why east, and why Manhattan is consistent

The agent moves east instead of north because A* uses the cost so far and the estimated distance to the goal to choose its next move. Since it assumes unknown cells are free, east appears to have a lower total estimated cost based on what the agent knows. Manhattan distance is consistent because moving one cell in any of the four directions changes the distance to the goal by exactly 1. Since each move costs 1, h(n) <= 1 + h(n') holds for neighboring cells, and h(goal) = 0.

Q2 Termination, and the bound on the number of moves

The agent finishes in finite time. Repeated Forward A* will eventually reach the target or determine it cannot be reached. Each time A* finds a path, the agent follows it until it reaches the goal or finds a blocked cell. When it finds a blocked cell, it updates the grid and plans again. Since the grid is finite, it cannot keep discovering new blocked cells forever. If n is the number of unblocked cells, each planned path has at most n moves. The agent can replan at most n times, giving a maximum of n^2 moves.

Q3 Tie-breaking: what you measured and why it happens

Breakaing ties towards larger g cut expanstions on both families, but much more on scatter. The maze using small g is 45,799, which is 3 times larger than when it uses the large g 15,058. The scatter map using small g is 44,696 which is many more times than 3,394. Because Manhattan distance is close to the real distance, small g goes through the whole region. Large g skips most of it. Walls also make the true distance longer than the Manhattan Distance.

Q4 Forward vs. backward: where the asymmetry comes from

The similarity between repeated forward and backward A* is that after the character moves, its position then changes, but the goal is still the same. Forward A* searches from the character to the goal using the new map after an obstacle is found. Backward A* searches from the goal to the character, and its solution is reversed. Both Repeated Forward and backward A* also use the new map to find a new path or route when they discover that the path they were first planning to use has an obstacle.

Q5 Adaptive A*, and whether the difference is real

Adaptive A* learns C-g(s) for expanded states, keeping estimates admissible and consistent even when costs rise. It helps more on mazes because walls cause detours Manhattan misses. Ratios were 0.51 for mazes and 0.96 for scatter. We tested no paired difference on the same maps. Maze p=7.79e-10 was significant, scatter p=0.00103 was not. Ratios show effect size; not significant doesn't mean equal. Wilcoxon needs independent maps and symmetric differences; paired t needs roughly normal differences.

Q6 Your differential heuristic, and when it fails to pay

I used the four grid corners as landmarks, choosing nearby cells when obstacles blocked them, and BFS to create distance tables. The differential heuristic is consistent because it does not overestimate. Older tables are admissible because discovering obstacles can increase path lengths. I compared the number of cells searched on the map and when obstacles forced the agent to replan. Although building and updating the tables takes extra time, they reduce the number of cells searched and improve efficiency.

Q7 Weighted A*: the bound, and why the loop erases the gain

Weighted A* guarantees a path cost within a factor of w of the optimal cost when using a consistent heuristic. Increasing the weight reduced expansions with a small increase in path cost, making it the best balance between speed and quality. However, obstacles can increase total search effort and travel distance. Adaptive A* should always use a weight of 1 because higher weights can make its learned heuristic inaccurate and break its guarantees.